

JS

JavaScript Mastery Roadmap

Beginner · Intermediate · Advanced

A comprehensive, grouped topic checklist for full-stack developers. Related topics organized together for efficient, connected learning.

71

Beginner

+17 new

89

Intermediate

+19 new

110

Advanced

+25 new

270

Total Topics

(+61 new topics added)



5-Star Rated Roadmap

NEW

= 61 newly added topics to fill knowledge gaps

How to Use This Roadmap

This roadmap organizes 209 original topics from your checklist plus 61 newly added topics into logical learning groups. Related topics are placed together so you can study connected concepts in one session — for example, Hoisting sits right next to Temporal Dead Zone, and Closures are grouped with Memoization and Partial Application.

Learning Tips

Study one group at a time. Check off topics as you master them. Topics marked with a **NEW** badge were added to fill knowledge gaps found during evaluation — they are equally important.

Evaluation Notes

The original checklist was excellent — well-structured and comprehensive. Key improvements made: (1) Topics regrouped by relatedness for connected learning, (2) Missing foundational topics added (truthy/falsy values, logical operators, Map/Set, event delegation), (3) Modern APIs and emerging standards added (Temporal API, Iterator helpers, import maps), (4) Practical browser APIs section added to Intermediate level, (5) 61 total new topics integrated across all levels.

BEGINNER

Foundation — Build your JavaScript fundamentals

71

topics (54 original + 17 new)

Variables & Data Types 10 topics

- ☐ 01 `var, let, const` — differences & scoping
- ☐ 02 Primitive types: `string, number, boolean, null, undefined, symbol, bigint`
- ☐ 03 `typeof` operator
- ☐ 04 Type coercion — implicit vs explicit
- ☐ 05 `==` vs `===` (loose vs strict equality)
- ☐ 06 `NaN, Infinity, -Infinity`
- ☐ 07 Truthy and falsy values **NEW**
- ☐ 08 Logical operators (`&&, ||, !`) **NEW**
- ☐ 09 Short-circuit evaluation **NEW**
- ☐ 10 Comparison operators (`<, >, <=, >=`) **NEW**

Strings 5 topics

- ☐ 11 String methods: length, indexOf, slice, split, join, trim, replace, includes
 - ☐ 12 Template literals (backtick strings)
 - ☐ 13 String immutability
 - ☐ 14 String.raw tagged template basics NEW
 - ☐ 15 String.prototype.at() for character access NEW
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Numbers & Math 4 topics

- ☐ 16 Number methods: toFixed, parseInt, parseFloat, Number()
 - ☐ 17 Math object: floor, ceil, round, random, max, min
 - ☐ 18 BigInt basics and when to use it NEW
 - ☐ 19 Number.isNaN() vs global isNaN() NEW
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Arrays 8 topics

- ☐ 20 Array creation and indexing
 - ☐ 21 Mutating methods: push, pop, shift, unshift, splice, slice
 - ☐ 22 Iterating methods: map, filter, reduce, forEach, find, findIndex, some, every
 - ☐ 23 Utility methods: flat, flatMap, includes, indexOf
 - ☐ 24 Array spread operator (...)
 - ☐ 25 Array destructuring
 - ☐ 26 Array.isArray() type checking NEW
 - ☐ 27 Array.from() and Array.of() NEW
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Objects 7 topics

- ☐ 28 Object creation: literal, new Object()
 - ☐ 29 Dot notation vs bracket notation
 - ☐ 30 Object.keys, Object.values, Object.entries, Object.assign
 - ☐ 31 Object spread operator
 - ☐ 32 Object destructuring with defaults and renaming
 - ☐ 33 Shorthand property names NEW
 - ☐ 34 Computed property names basics NEW
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Control Flow & Conditionals 5 topics

- ☐ 35 if / else if / else
 - ☐ 36 switch / case / default
 - ☐ 37 Ternary operator
 - ☐ 38 Nullish coalescing operator (??)
 - ☐ 39 Optional chaining (?.)
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Loops & Iteration 4 topics

- ☐ 40 for loop, while loop, do...while
 - ☐ 41 for...of — arrays and iterables
 - ☐ 42 for...in — object keys
 - ☐ 43 break, continue, labeled statements
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Functions 8 topics

- ☐ 44 Function declaration vs function expression
 - ☐ 45 Arrow functions (=>)
 - ☐ 46 Default parameters
 - ☐ 47 Rest parameters (...args)
 - ☐ 48 Arguments object
 - ☐ 49 Return values
 - ☐ 50 IIFE (Immediately Invoked Function Expression)
 - ☐ 51 Callback functions basics NEW
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Scope & Hoisting 4 topics

- ☐ 52 Global scope, function scope, block scope
 - ☐ 53 Hoisting — var, let, const, function declarations
 - ☐ 54 Temporal Dead Zone (TDZ)
 - ☐ 55 Strict mode ('use strict') NEW
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Error Handling Basics 3 topics

- ☐ 56 try / catch / finally
- ☐ 57 throw new Error()
- ☐ 58 Error types: TypeError, ReferenceError, SyntaxError, RangeError

DOM Manipulation & Events 7 topics

- ☐ 59 document.querySelector, querySelectorAll
- ☐ 60 getElementById, getElementsByClassName
- ☐ 61 innerHTML, textContent, innerText
- ☐ 62 setAttribute, getAttribute, classList
- ☐ 63 addEventListener, removeEventListener
- ☐ 64 Event object: target, preventDefault, stopPropagation
- ☐ 65 Event bubbling & capturing basics NEW

Built-in Utilities 6 topics

- ☐ 66 console.log, console.error, console.warn, console.table
- ☐ 67 JSON.stringify, JSON.parse
- ☐ 68 setTimeout, setInterval, clearTimeout, clearInterval
- ☐ 69 Date object basics
- ☐ 70 Comments: single-line (//) and multi-line (/* */) NEW
- ☐ 71 console.dir, console.group, console.time NEW

INTERMEDIATE

Real-world code — Patterns & professional skills

89

topics (70 original + 19 new)

Closures & Lexical Scope 5 topics

- ☐ 01 Closures — lexical scope & practical use cases
 - ☐ 02 Closure-based private state (module pattern)
 - ☐ 03 Memoization with closures
 - ☐ 04 Partial application with closures
 - ☐ 05 Variable shadowing
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Prototypes & Classes 10 topics

- ☐ 06 Prototype chain — `__proto__` & `Object.prototype`
 - ☐ 07 `Object.create(proto)`
 - ☐ 08 Constructor functions & `new` keyword
 - ☐ 09 class syntax — constructor, methods, getters/setters
 - ☐ 10 `extends` & `super` keyword
 - ☐ 11 static methods & properties
 - ☐ 12 Private class fields (`#`)
 - ☐ 13 `instanceof` operator
 - ☐ 14 `hasOwnProperty` vs `Object.hasOwnProperty()`
 - ☐ 15 Mixins pattern with classes NEW
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Property Descriptors & Object Control 4 topics

- ☐ 16 Property descriptors: writable, enumerable, configurable
 - ☐ 17 `Object.defineProperty`, `Object.freeze`, `Object.seal`
 - ☐ 18 `Object.preventExtensions()` NEW
 - ☐ 19 Enumerability and property iteration NEW
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this Keyword & Binding 4 topics

- ☐ 20 `this` — 5 binding rules (default, implicit, explicit, new, arrow)
 - ☐ 21 `call()`, `apply()`, `bind()`
 - ☐ 22 Arrow function & `this` (lexical binding)
 - ☐ 23 `this` in callbacks & event handlers
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Asynchronous JavaScript 11 topics

- ☐ 24 Callback functions & callback hell
 - ☐ 25 Promise — states (pending, fulfilled, rejected)
 - ☐ 26 Promise chaining (.then, .catch, .finally)
 - ☐ 27 Promise.all, Promise.allSettled, Promise.race, Promise.any
 - ☐ 28 async / await syntax
 - ☐ 29 Error handling with try/catch in async functions
 - ☐ 30 Floating promises (unhandled rejections)
 - ☐ 31 async/await in loops — sequential vs parallel
 - ☐ 32 Promise.withResolvers()
 - ☐ 33 Top-level await NEW
 - ☐ 34 Async error patterns: fail-fast vs fail-safe NEW
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Modules & Module Systems 5 topics

- ☐ 35 ES Modules: import, export, export default
 - ☐ 36 Named exports vs default exports
 - ☐ 37 Dynamic import() — lazy loading
 - ☐ 38 CommonJS: require, module.exports
 - ☐ 39 CommonJS vs ES Modules differences
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Advanced Destructuring & Syntax 6 topics

- ☐ 40 Nested destructuring
 - ☐ 41 Computed property names in destructuring
 - ☐ 42 Swap variables with destructuring
 - ☐ 43 Function parameter destructuring
 - ☐ 44 Logical assignment operators (??=, ||=, &&=) NEW
 - ☐ 45 Optional catch binding NEW
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Iterators & Generators 5 topics

- ☐ 46 Iterator protocol (`next()`, `{value, done}`)
 - ☐ 47 Generator functions (`function*`) & `yield`
 - ☐ 48 Infinite sequences with generators
 - ☐ 49 `for...of` with custom iterables
 - ☐ 50 `Symbol.iterator`
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Collections: Map, Set & Weak References 4 topics

- ☐ 51 Map and Set data structures NEW
 - ☐ 52 Map vs Object — when to use which NEW
 - ☐ 53 WeakMap & WeakSet
 - ☐ 54 WeakRef & FinalizationRegistry
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Symbols & Metaprogramming Basics 4 topics

- ☐ 55 Symbol — unique keys, `Symbol.for`, `Symbol.keyFor`
 - ☐ 56 Well-known symbols: `Symbol.toPrimitive`, `Symbol.hasInstance`, `Symbol.species`
 - ☐ 57 Proxy & Reflect — traps (`get`, `set`, `apply`)
 - ☐ 58 Tagged template literals
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Modern ES Features (ES2022–2024) 8 topics

- ☐ 59 `Object.groupBy` (ES2024)
 - ☐ 60 `Array.at(-1)` negative indexing
 - ☐ 61 `Object.hasOwn()`
 - ☐ 62 Error cause: `new Error('msg', {cause: err})`
 - ☐ 63 `structuredClone()`
 - ☐ 64 Intl API basics (`NumberFormat`, `DateTimeFormat`) NEW
 - ☐ 65 `Array.findLast()` and `Array.findLastIndex()` NEW
 - ☐ 66 `String.prototype.replaceAll()` NEW
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Functional Programming Patterns 7 topics

- ☐ 67 Pure functions & side effects
 - ☐ 68 Immutability patterns (spread, Object.freeze)
 - ☐ 69 Higher-order functions
 - ☐ 70 Function composition — compose & pipe
 - ☐ 71 Currying & partial application
 - ☐ 72 Array method chaining
 - ☐ 73 Transducers
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Regular Expressions 6 topics

- ☐ 74 RegEx literals & RegExp constructor
 - ☐ 75 test(), match(), matchAll(), replace(), search(), split()
 - ☐ 76 Character classes, quantifiers, groups
 - ☐ 77 Lookahead & lookbehind assertions
 - ☐ 78 Named capture groups
 - ☐ 79 Unicode-aware regex with /u flag NEW
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Advanced Error Handling 5 topics

- ☐ 80 Custom Error classes
 - ☐ 81 Error chaining (cause)
 - ☐ 82 Unhandled promise rejection — process.on('unhandledRejection')
 - ☐ 83 Global error handlers
 - ☐ 84 AggregateError for multiple errors NEW
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Browser APIs & Patterns 5 topics

- ☐ 85 Event delegation pattern NEW
 - ☐ 86 Fetch API & Response handling NEW
 - ☐ 87 Local Storage & Session Storage NEW
 - ☐ 88 History API basics (pushState, replaceState) NEW
 - ☐ 89 URL and URLSearchParams NEW
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ADVANCED

Senior developer level — Deep internals & architecture

110

topics (85 original + 25 new)

V8 Engine Internals 6 topics

- ☐ 01 V8 engine — parsing, Ignition bytecode, Turbofan JIT
- ☐ 02 Hidden classes & inline caches
- ☐ 03 Monomorphic vs polymorphic functions
- ☐ 04 Deoptimization triggers
- ☐ 05 Object shape consistency for V8 optimization
- ☐ 06 Abstract Syntax Tree (AST) basics NEW

Memory Management & Garbage Collection 5 topics

- ☐ 07 JavaScript heap — new space (young gen) vs old space (old gen)
- ☐ 08 Mark-and-sweep garbage collection
- ☐ 09 Generational GC — minor GC vs major GC
- ☐ 10 Memory leaks: global vars, event listeners, detached DOM, closures
- ☐ 11 Chrome DevTools — Memory tab, heap snapshot, allocation timeline

Event Loop & Task Scheduling 9 topics

- ☐ 12 Call stack, Web APIs, callback queue, microtask queue
- ☐ 13 Microtask queue vs macrotask queue — execution order
- ☐ 14 Node.js event loop phases (timers, I/O, poll, check, close)
- ☐ 15 process.nextTick vs Promise microtask — priority
- ☐ 16 scheduler.yield() — yielding to the event loop
- ☐ 17 requestAnimationFrame & requestIdleCallback
- ☐ 18 Long tasks & main thread blocking
- ☐ 19 Event loop lag monitoring
- ☐ 20 queueMicrotask() API NEW

Execution Context & Scope Internals 5 topics

- ☐ 21 Execution context — creation phase vs execution phase
 - ☐ 22 Variable Environment vs Lexical Environment
 - ☐ 23 Scope chain resolution
 - ☐ 24 Call stack frames
 - ☐ 25 Tail call optimization (TCO) concept NEW
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Advanced Async Patterns 9 topics

- ☐ 26 AbortController & AbortSignal — request cancellation
 - ☐ 27 Async generator functions (async function*)
 - ☐ 28 for await...of — async iteration
 - ☐ 29 AsyncLocalStorage (Node.js) — request context propagation
 - ☐ 30 Concurrency limiter — process N items with max K parallel
 - ☐ 31 Retry with exponential backoff
 - ☐ 32 Circuit breaker pattern
 - ☐ 33 Promise queue
 - ☐ 34 Async resource management (using declaration) NEW
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Streams & Data Flow 4 topics

- ☐ 35 Readable, Writable, Transform, Duplex streams
 - ☐ 36 Backpressure & flow control
 - ☐ 37 pipeline() — error-safe stream chaining
 - ☐ 38 Web Streams API — ReadableStream, WritableStream
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Multi-threading & Concurrency 7 topics

- ☐ 39 Worker Threads vs `child_process`
 - ☐ 40 `SharedArrayBuffer` & `Atomics`
 - ☐ 41 Transferable objects – zero-copy transfer
 - ☐ 42 Worker pool pattern
 - ☐ 43 Comlink – wrap workers as async proxies
 - ☐ 44 Service Workers lifecycle NEW
 - ☐ 45 Web Workers communication patterns NEW
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Proxy, Reflect & Metaprogramming 6 topics

- ☐ 46 Proxy – all traps: `get`, `set`, `has`, `deleteProperty`, `apply`, `construct`
 - ☐ 47 Reflect – Proxy traps mirror
 - ☐ 48 Revocable proxy
 - ☐ 49 TC39 Decorators (Stage 3) – `@logged`, `@memoize`, `@readonly`
 - ☐ 50 `Symbol.asyncIterator` – async iteration protocol
 - ☐ 51 `Symbol.toPrimitive` deep dive
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Functional Programming – Advanced 6 topics

- ☐ 52 Maybe monad – null-safe chaining
 - ☐ 53 Either monad – error handling without exceptions
 - ☐ 54 Functor & `flatMap` (`bind`)
 - ☐ 55 Lazy evaluation with generators
 - ☐ 56 Point-free programming style
 - ☐ 57 Algebraic data types in JS NEW
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Design Patterns 12 topics

- ☐ 58 Singleton pattern
 - ☐ 59 Factory & Abstract Factory
 - ☐ 60 Builder pattern
 - ☐ 61 Observer / EventEmitter pattern
 - ☐ 62 Strategy pattern
 - ☐ 63 Command pattern (undo/redo)
 - ☐ 64 Proxy pattern
 - ☐ 65 Decorator pattern
 - ☐ 66 State machine pattern
 - ☐ 67 Module pattern (IIFE-based)
 - ☐ 68 Mediator / Middleware pattern **NEW**
 - ☐ 69 Repository pattern **NEW**
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Performance Optimization 11 topics

- ☐ 70 Bundle analysis — webpack-bundle-analyser
 - ☐ 71 Tree shaking — ES modules & dead code elimination
 - ☐ 72 Code splitting & dynamic import()
 - ☐ 73 Debounce & throttle — from scratch
 - ☐ 74 Virtual scrolling concept
 - ☐ 75 Performance.now() — micro-benchmarking
 - ☐ 76 Typed arrays — Int32Array, Float64Array, performance benefits
 - ☐ 77 Intersection Observer API **NEW**
 - ☐ 78 Mutation Observer API **NEW**
 - ☐ 79 Performance API: PerformanceObserver, marks, measures **NEW**
 - ☐ 80 Lazy loading images & resources **NEW**
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Security 8 topics

- ☐ 81 XSS — innerHTML danger, textContent safety, DOMPurify
 - ☐ 82 Content Security Policy (CSP)
 - ☐ 83 Prototype pollution attack & defense
 - ☐ 84 eval() & new Function() — security risks
 - ☐ 85 HttpOnly cookies & JavaScript access
 - ☐ 86 Supply chain security — npm audit, lockfile
 - ☐ 87 CSRF protection in JavaScript NEW
 - ☐ 88 Subresource Integrity (SRI) NEW
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Type Safety & Documentation 4 topics

- ☐ 89 JSDoc type annotations
 - ☐ 90 Type narrowing with typeof, instanceof, in
 - ☐ 91 Discriminated unions pattern in JS
 - ☐ 92 Runtime type validation (Zod, Yup concepts) NEW
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Testing 6 topics

- ☐ 93 Unit testing mindset — pure functions first
 - ☐ 94 Mocking — stubs, spies, mocks differences
 - ☐ 95 Property-based testing concept
 - ☐ 96 TDD — red/green/refactor cycle
 - ☐ 97 Integration testing strategies NEW
 - ☐ 98 Test doubles: fake, stub, spy, mock, dummy NEW
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Node.js Internals & Tooling 7 topics

- ☐ 99 Node.js module system — CJS caching, circular deps
 - ☐ 100 Dual ESM/CJS package publishing
 - ☐ 101 UV_THREADPOOL_SIZE — libuv thread pool tuning
 - ☐ 102 Building a CLI tool — Commander.js, Inquirer.js
 - ☐ 103 Publishing an npm package — semver, package.json bin field
 - ☐ 104 Node.js built-in test runner NEW
 - ☐ 105 Import maps & module resolution NEW
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Emerging Standards & Proposals 5 topics

- ☐ 106 Temporal API (date/time replacement) NEW
 - ☐ 107 Record & Tuple proposal (immutable data) NEW
 - ☐ 108 Pattern matching proposal NEW
 - ☐ 109 ShadowRealm API concept NEW
 - ☐ 110 Iterator helpers (map, filter, take on iterators) NEW
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Your JavaScript Journey

270 topics across 3 levels, organized into 43 learning groups.

Check them off one by one. You've got this.